Telegram Wallpaper Bot

This is a Telegram bot built with `python-telegram-bot` and `MongoDB Atlas` that allows users to search, browse, and request wallpapers. The bot owner can upload new wallpapers and view statistics.

Features

1. **UPLOAD (Owner only)**: Owner sends a wallpaper image/file to the bot, then the bot asks for tags. The wallpaper is stored with those tags. Multiple tags per wallpaper supported.
2. **SEARCH**: Users type a character/anime name (e.g., "naruto") or use `/search <name>` and the bot sends all matching wallpapers. Search matches against tags (case-insensitive and partial match).
3. **BROWSE BY CATEGORY**: Show inline keyboard buttons for categories: Naruto, One Piece, Jujutsu Kaisen, Dragon Ball, Demon Slayer, Attack on Titan, My Hero Academia. Tapping a category shows wallpapers in that category.
4. **RANDOM**: `/random` gives a random wallpaper. `/random <tag>` gives a random wallpaper tagged with the specified tag.
5. **POPULAR/TRENDING**: Tracks how many times each wallpaper is requested/viewed. `/trending` shows the most popular wallpapers.
6. **REQUEST SYSTEM**: Users can use `/request <description>` (e.g., `/request gojo purple wallpaper`). The owner gets notified of the request.
7. **DAILY WALLPAPER**: Users can subscribe with `/daily`. Every day at 12:00 PM IST, subscribed users get a random wallpaper. `/stopdaily` to unsubscribe.
8. **STATS (Owner only)**: `/stats` shows total users, total wallpapers, most searched terms, total downloads/views.

Commands

- `/start` — Welcome message with instructions
- `/help` — Show all commands
- `/search <name>` — Search wallpapers (or just type the name directly)
- `/browse` — Show category buttons
- `/random` — Random wallpaper
- `/random <tag>` — Random from specific tag

- `/trending` — Most popular wallpapers
- `/request <description>` — Request a wallpaper
- `/daily` — Subscribe to daily wallpaper
- `/stopdaily` — Unsubscribe from daily
- `/stats` — Owner only stats
- `/upload` — Owner only, start upload mode

Setup Instructions

1. Obtain Bot Token and Owner ID

- **Bot Token**: Talk to BotFather on Telegram to create a new bot and get your bot token.
- **Owner Telegram ID**: Forward any message from yourself to the `@userinfobot` on Telegram to get your user ID. This ID will be used to grant you owner-only permissions for upload and stats.

2. Set up MongoDB Atlas

1. Go to MongoDB Atlas and create a free tier account.
2. Create a new cluster.
3. Create a database user with a strong password.
4. Set up network access to allow connections from anywhere (or specific IPs if you know them).
5. Get the connection string for your cluster. It will look something like `mongodb+srv://<username>:<password>@<cluster-name>.mongodb.net/?retryWrites=true&w=majority`.

3. Environment Variables

Create a `.env` file in the root directory of your project with the following variables:

Plain Text

```
BOT_TOKEN=YOUR_BOT_TOKEN
OWNER_TELEGRAM_ID=YOUR_TELEGRAM_USER_ID
MONGODB_CONNECTION_STRING="YOUR_MONGODB_ATLAS_CONNECTION_STRING"
```

Replace `YOUR_BOT_TOKEN`, `YOUR_TELEGRAM_USER_ID`, and `YOUR_MONGODB_ATLAS_CONNECTION_STRING` with your actual values.

4. Deployment on Railway

This bot is configured for deployment on Railway using Nixpacks.

1. **Create a new project on Railway.**
2. **Connect your GitHub repository** where this bot's code is hosted.
3. **Add Environment Variables**: In your Railway project settings, go to the "Variables" tab and add the `BOT_TOKEN`, `OWNER_TELEGRAM_ID`, and `MONGODB_CONNECTION_STRING` as environment variables. Make sure the names match exactly.
4. **Nixpacks Configuration**: The `nixpacks.toml` file is already included to ensure Python 3.11 and `gcc` are available for `APScheduler`.
5. **Procfile**: The `Procfile` specifies how to start the bot: `web: python3 bot.py`.
6. **Deploy**: Railway should automatically detect the `Procfile` and `nixpacks.toml` and deploy your bot. If not, manually trigger a deployment.

5. Running Locally (for development)

1. **Clone the repository:**

   Bash

   ```bash
   git clone <your-repo-url>
   cd telegram-wallpaper-bot
   ```

2. **Install dependencies:**

   Bash

   ```bash
   pip install -r requirements.txt
   ```

3. **Create a `.env` file** as described in step 3.
4. **Run the bot:**

   Bash

   ```bash
   python3 bot.py
   ```

Project Structure

Plain Text

```
.
├── bot.py
├── requirements.txt
├── nixpacks.toml
├── Procfile
└── README.md
```

Technologies Used

- Python 3.11
- `python-telegram-bot` library
- `MongoDB Atlas` (via `motor` and `pymongo`)
- `APScheduler` for daily tasks
- `dotenv-parser` for environment variable management
- Railway for deployment